CHAPTER FIVE

DEVELOPING THE TEACHING OF MATHEMATICS THROUGH A PROBLEM SOLVING APPROACH: REFLECTIONS AND PROJECTIONS

5.0 INTRODUCTION

This final chapter pulls together ideas on the development of the teaching of mathematics through a problem solving approach and makes projections for the implementation of this approach in Guyanese schools. The areas that will be discussed include:

- Problems of a problem solving approach
- Benefits of a problem solving approach
- Suggestions for teaching through a problem solving approach
- Ways in which a problem solving approach will be implemented in Guyana.

5.1 PROBLEMS OF A PROBLEM SOLVING APPROACH

For teachers to adapt to a problem solving approach, they need to make significant adjustments to their previous teaching behaviours. However, this cannot be accomplished without foresight of problems.

- *The nature of the learner's experience.* It is reasonable to assume that teaching mathematics via a problem solving approach will prove more demanding and less predictable for students than a traditional approach. This is mainly due to the fact, as pointed out by Blum (et al, 1991, p. 54), that: "Traditional routine tasks such as calculations are more popular with many students because they are much easier to grasp and can often be solved

merely by following certain recipes, which makes it easier for students to obtain good marks in tests and examinations". Consequently, as argued by Kaput (1989) and Mason (1988), I have noticed that many students act as though they are convinced that there is a simple translation process that moves someone from a problem to a mode of solution, (for example, the word problems to the algebra) without intervening thinking.

In addition, from my experience I am inclined to agree with Mouly (1973, p. 373) who claims that: "Much depends on whether the problem and the relationship among the parts which they must understand to reach a solution are within their understanding and experience". And further add, that students often fail to tackle mathematical problems because of their inadequate problem solving strategies. Therefore, one may conclude that students are relatively blind to the existence of problems because of their limited experience.

- *Restricted mathematical thinking.* Like the structure of many mathematics textbooks, teachers tend to break things down into smaller chunks in an attempt to make work accessible to students. Such action is likely to fail in helping pupils to think mathematically. In support of this view, Mouly (1973, p. 374) makes the point that: "In their eagerness to help children over every hump, teachers often deprive them of the opportunity to do any real thinking. Many lessons are so thoroughly premasticated or presented in such mechanical fashion that the child is left with nothing to do but implement on a prescription basis the teachers' solutions". Nevertheless, it is worth noting that many teachers were themselves the product of traditional mathematics education and hence, their ability to think mathematically is questionable.

- *The impact of students' and teachers' beliefs.* "Teachers are people and people in general are resistant to imposed changes; mathematics teachers in particular are resistant to imposed changes in their approach to teaching. When new ideas conflict with the beliefs of the individual, then the new ideas are either not heard at all or not retained" (Mason, et al, 1991, p. 53). Both teachers' and students' disposition and beliefs are major obstacles to making significant changes in the teaching and learning of mathematics in schools. Nevertheless, teachers tend to change themselves with an awareness of new possibilities. In light of this, NCTM (1991, p. 1) claims that: "In order to change our perception about mathematics teaching and learning, we need direction on how mathematics can be taught and learned to enhance the development of mathematical power (problem solving ability)".

- *Interaction within the classroom.* Generally, it may be difficult to get pupils used to different patterns of discourse (NCTM, 1991). Pupils may feel more comfortable getting a response from the teacher like, "that is right" rather than "that is a good effort" since many of them prefer to know whether their solution is correct or incorrect. Further, from my own teaching experience and findings from an interview with the teacher of lesson D (Chapter three, page 42), I am forced to conclude that students do not readily coordinate their efforts in the process of problem solving. Nevertheless, teachers can play a significant role in securing efficient and effective verbal and social interaction within their classrooms to facilitate cooperative group work.

Finally, in spite of the possible problems outlined above, there are reasonable arguments in favour of problem solving. Sadly, problem solving

does not play as important a role in mainstream mathematics instructions at school and university as we would wish (Burkhardt, 1983). This may be due to many contributory factors such as; the teacher's perception on the nature of problem solving, the fear of time constraints to deal with problems solving and [as indicated by Blum (et al, 1991)] references to the world outside mathematics make instruction more demanding for teachers because additional "non-mathematical" qualifications are necessary and make it more difficult to assess students' achievement.

5.2 BENEFITS OF A PROBLEM SOLVING APPROACH

In Chapter two (page 24), it was said that mathematics educators have viewed problem solving as a preferential objective of construction. Therefore, despite the constraints of problem solving mentioned in the previous section there must be significant benefits to be gained for the learner from the teaching of mathematics via a problem solving approach. Some of these benefits are outlined below.

- *Developing creative talents.* "I am asked to be adaptable in different situations, to see that different situations require different questions to work on them" (Hewitt, 1992, p. 8). This statement, in a sense, projects that mathematics teachers should capitalise on opportunities to identify and develop children's creative talents. In light of this, as stated in Chapter two, Cangelosi (1992, p. 298) claims that: "Virtually everyone needs to be able to use mathematics creatively to solve real-life problems". Further, Torrance (1966) adds that virtually everyone possess creative talents but what is rare is for that talent to be nurtured. Teachers should nurture this talent if they are to help the student to avoid losing their inborn facility. However, the probability of developing children's creative ability may increase if they are

given the opportunity to be engaged in what Woods (1988, p. 9) calls "real problem solving". That is: "the opportunity to make their (pupils) own decisions, take responsibility for themselves and cope with the consequences".

- *Improved teacher awareness.* Too often mathematics problems are restricted to problems within mathematics. By teaching mathematics through problem solving we have the opportunity of demonstrating the relevance of mathematical knowledge and skills outside mathematics. In a sense, as pointed out by Brown (et al, 1978, p. 391), "teachers can integrate branches of mathematics (and/or other subjects) and extend their command of the subject through problem solving". Further, teachers may tend to "empathise with the difficulties many students have in problem solving and thus, may gain greater respect for the student's mind" (Brown, et al, ibid).

- *Enhanced mathematical empowerment.* Basically, as Bell (et al, 1978) points out, problem solving enables pupils to exercise and acquire confidence in their intellectual power. This view was echoed by Woods (1988, p. 9) who said: "My experience of real problem solving has shown that it is an approach which has a great deal to offer pupils whose confidence is boosted by the chance of really doing something themselves as well as by the opportunity to make real decisions". Consequently, pupils are likely to gain confidence in using mathematics as a tool in their everyday living - which should be one of the main goals of teaching mathematics.

One of the roles of schools is to prepare students for 'life'. Therefore, any attempt to develop students' analytical and reasoning skills must be welcomed. According to Broomes (et al, 1993, p. 2) "Problem solving assists

pupils in developing analytical and reasoning skills". In the process of problem solving students will usually have to demonstrate some level of inventiveness. The relatively new and challenging tasks, pupils are provided with, have the tendency to force pupils to evaluate and modify their thinking processes whenever new information becomes available and probably devise their own method of attacking problems.

Finally, during the process of problem solving there may be the tendency for the solver to ponder and reflect on their solutions especially if they are engaged in cooperative group work. In other words, pupils may be exposed to the art of looking back. As mentioned in Chapter two (page 30), the merits of 'looking back' cannot be over-emphasised. Also, problem solving offers opportunities to help pupils develop both understanding of concepts and facility with procedure, and to apply and review and make connections with processes already learned - as evident in ***Case IIc*** (page 39).

5.3 SUGGESTIONS FOR TEACHING THROUGH A PROBLEM SOLVING APPROACH

Having considered some relevant literature in Chapter two, presented and analysed some anecdotes from mathematics lessons in Chapters three and four respectively, and anticipated problems and benefits of adopting a problem solving approach to the teaching of mathematics; I will offer the following suggestions which I consider applicable to any teacher who is genuinely concerned about developing their teaching techniques.

- *Capitalise on collegiality.* Much can be gained by having discussion and collaborating with colleagues on the teaching and learning of mathematics especially via a problem solving approach. In light of this,

Cangelosi (1992) claims that fellow mathematics teachers within your schools and even those at nearby schools may be the most immediate and sympathetic resources for suggestions and ideas on improving your mathematics teaching. Therefore, innovative teachers should visit each others classrooms (Allen, et al, 1984), compare notes and arrive at a consensus on teaching techniques after discussing particular aspects of lessons. Further, sharing responsibilities for students' learning (Cangelosi, 1988a) is also useful.

- *Managing classrooms wisely.* Orton (1994, p. 165) states that: "The first change with teachers' experience when introducing exploration of problems (problem solving) involves a loosening of their reins of control". Students should feel free to and begin to make their own decisions. Initially as students begin to make their own decisions, teachers may find it difficult to cope with this new student behaviour and experience a sense of insecurity. However, in the interests of students constructing their own knowledge and the merits of engaging pupils in the process of problem solving, students should be allowed to enjoy their newly found freedom which may lead to them working together, cooperatively in groups.

Further, it is expected that students may initially be reluctant to express their ideas to the whole class. In this case, discussions in pairs or small groups may give students some confidence. Discussions are important because, among other things, as Wood (1988) pointed out, a trouble shared, in mathematical discussion, may become a problem solved.

- *Enhanced teacher awareness.* Teachers who choose to adopt a problem solving approach to their teaching should, themselves, have

encountered the process of problem solving. In light of this Brown (et al, 1978, p. 391) postulates that: "As the teacher develops his own problem solving heuristics, he may find better ways of communicating similar problem processing strategies to his students". Moreover, it is important for the teacher to experience the pleasure and frustrations which are involved in such an approach to the study of mathematics if they are to improve their classroom effectiveness.

On the other hand, although the teacher may have gained insight into ideas, which they may share with their students, having explored problems beforehand such action(s) have limitations. Knowing what path(s) are most effective to a solution the teacher may have the tendency to propose how and what route students should pursue in solving a particular problem.

The ingenious teacher who wishes to introduce the problem solving approach should not only ensure that the mathematical problems reflect situations that are reasonably difficult to be challenging, but reasonably simple and familiar for pupils to manage. Therefore, whenever possible, students should have opportunities to formulate problems and questions that originate from their own interest. Such action may result in a sense of ownership which can lead to some level of motivation. "Motivation makes for greater persistence and for the development of a more adequate background, both of which facilitate problem solving" (Mouly, 1973, p. 377).

In addition, as suggested in Chapter four, the process of problem solving is greatly enhanced with improved ability to form imagery, conjecture, specialise and generalise. Further, teachers should [as mentioned in Chapter two (page 30)] have all students engaged in and exposed to the art

of 'looking back' to enable them to extend their knowledge. In this vein Mouly (1973, p. 373) claims that: "The ability to extend the applicability of a solution from one situation to another is the real test of understanding; it is also the basis for scientific progress".

- *Maximising the use of educational technology.* Educational technology; calculators, computers and videos should be used when appropriate. Computers, and calculators to some extent, free students from tedious computation and allow them to concentrate on the nature of the content and the process of problem solving. In light of this, we are reminded by Blum (1991, p. 52) that "computers (and possibly calculators) allow for greater detail, accuracy and rapidity in calculations, for the handling of more data, for examining the effects of changing terms or parameters, for providing better illustrations". Further, computers and calculators can be seen as personal technology. When pupils are engaged in individual work they may develop confidence in their own ability to solve problems since they are allowed to make their mistakes privately and get immediate feedback.

On the other hand, videos may prove to be excellent in introducing and sustaining interest in problem solving and provide a strong cross-curriculum link and among subjects such as mathematics, science, geography and social studies. The advantage of using the videos is not only that most students are interested in television and videos but videotapes can be interrupted, replayed and/or restarted on demand without loss of words. Therefore, routine classroom activities such as comments and questions do not interfere with the line of thought on the video. Hints on engaging pupils in productive discussion after viewing a videotape can be had from appendix III (page 98).

- *Re-examine your beliefs.* In Chapter four (page 58), I said that teachers' beliefs about mathematics can shape their teaching styles. Many of our beliefs, such as the way in which pupils learn mathematics and the nature of problem solving, are manifestations of our own experiences as students. We need to rethink and re-examine these beliefs; in the interest of improving our mathematics teaching techniques. However, since we are the only ones who can change or adjust our beliefs, it is desirable to challenge them by: having discussions with colleagues; reading current literature; attending workshops and/or other staff development sessions; and, if possible, attending training programmes on the nature of learning and problem solving.

- *View assessment as a continuum.* Assessment, which involves the collecting and interpreting of information to aid decision making (Airasan, 1991), is of paramount importance. In attempting to improve our teaching of mathematics much emphasis must be given the to assessment of pupils' mathematical achievement. In light of this, Lincoln and Guba (1981) indicated that assessment is concerned with *refinement* and *improvement.* Teachers should be constantly assessing students' progress if they are to provide suitable prescription in attempting to foster children's problem solving ability. In the lessons outlined in Chapter three (especially lesson D) the teachers were constantly assessing students' progress.

In the process of problem solving the teacher should be making assessment of pupils' progress throughout the particular activity. Sometimes it is difficult for an 'outsider' to determine whether the pupils are learning since it is only the particular teacher who knows what they are trying to do. However, possible areas of assessment (adapted from Cangelosi, 1992, p. 320)

should include: mathematical empowerment; "the ability to analyse and reason, and apply their knowledge to solve problems and other subjects, apply various methods to solve problems, verify and interpret results, and generalise solutions". Also, assessment techniques should include having students react to and evaluate another student's process for solving a particular problem. "Pupils confidence in using mathematics to solve problems, to communicate ideas and to reason, and their flexibility in exploring mathematics ideas and trying alternate methods in problem solving" should also be considered (Cangelosi, ibid, p. 320).

Above all, students need to carry out problem solving without any threat of assessment. This view was supported by Isaacs (1987) who claimed that most students should be allowed to demonstrate their problem solving skills in a relaxed environment and extended time period - as in projects.

Finally, if teachers are to follow the problem solving approach, they need to foster in their students a critical attitude; a willingness to say "why" rather than simply "what". Above all, in implementing the problem solving approach to the teaching of mathematics, the teacher must perform the roles of creator, facilitator, intervener and act as a model. Description of these roles can be had from appendix IV (page 100).

5.4 WAYS IN WHICH A PROBLEM SOLVING APPROACH WILL BE IMPLEMENTED IN GUYANA

At the beginning of Chapter one, I said that, my main roles in Guyana as lecturer in the Faculty of Education include:

- giving instruction in pedagogy to mathematics teachers from primary, secondary and tertiary (including university) institutions